Your 100MB graph is loaded correctly.

Current counts

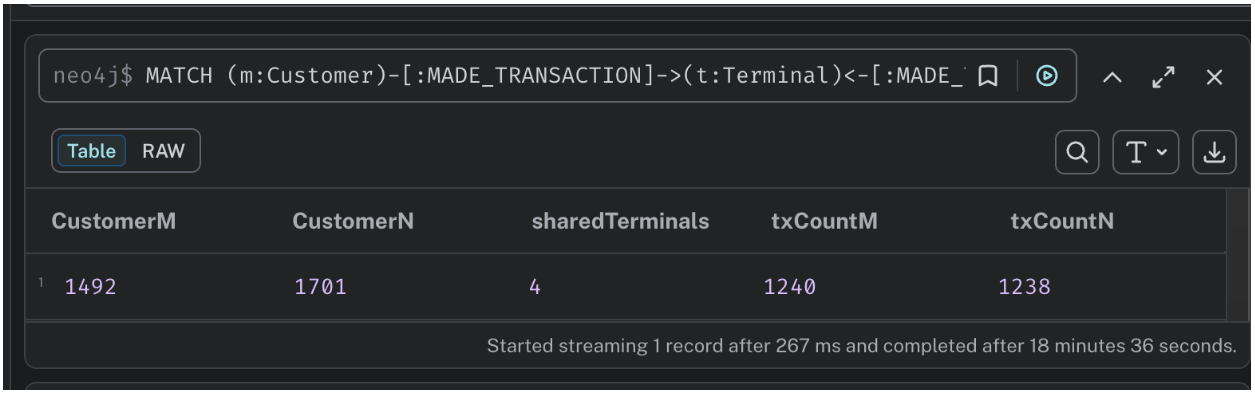
Entity	Count
Customers	2,000
Terminals	300
Transactions	2,031,891

Import timings

Operation	Time
Customer import	212 ms
Terminal import	31 ms
Transaction import	12.4 s

DATASET	QUERYA	QUERYB	QUERYD1	QUERY D2	QUERYE
50MB	4 min 46 sec			7 min 28 sec	4.650 sec
100MB	18 min 36 sec	OUT OF MEM		30 min 53 sec	8.888 sec
200MB					

Query 1 :



Query A exhibits super-linear growth. Doubling the transaction volume from 50MB to 100MB increased execution time by almost four times, indicating that the query is highly sensitive to graph density and customer-to-customer comparison costs.

$18.6 / 4.77 \approx 3.9\times$

Query A

50MB:

Result: Customer 861 ↔ 1228

Shared terminals: 4

Time: 4m 46s

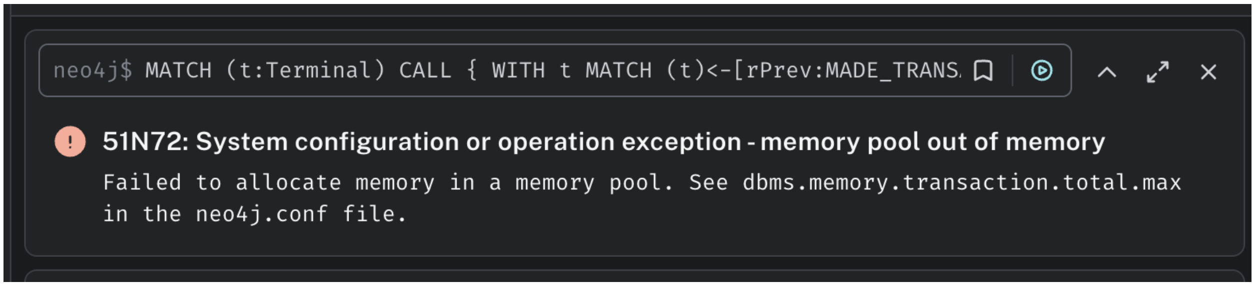
100MB:

Result: Customer 1492 ↔ 1701

Shared terminals: 4

Time: 18m 36s

QUERY B.



The query executed successfully on the 50MB dataset but exhausted available memory on the 100MB dataset, demonstrating poor scalability.

QUERY D.2

SIMILAR_BEHAVIOR relationships created: 33,154

Properties set: 66,308

Execution time: 30 min 53 sec

Interesting observation:

- The number of resulting relationships barely increased:
    - 32,323 → 33,154 (+2.6%)
- But execution time increased:
    - 7m 28s → 30m 53s (~4.1×)

This is a strong discussion point for your report:

Although the resulting graph structure remained similar, the computational cost of discovering similar customer pairs increased substantially as transaction volume grew.